Submission Overview

A read-only QuickBooks connection, a monthly financial-analysis skill, an evaluation harness and a one-page architecture note, built for the RE Financial Services build exercise. Python computes every number; Claude writes the commentary; a second, read-only Claude agent and a deterministic grader check it; a human signs off.

What was built

Part	What exists and works today
1. Connection	Intuit's own `quickbooks-online-mcp-server`, run locally over stdio against the Intuit developer **sandbox**, with the write, update and delete flags set to `true` (disabled), so no write tool is registered. Live read-only checks passed on 2026-10-14: CompanyInfo, a monthly P&L (Jan–Jun 2026), a Balance Sheet at 30 Jun 2026 (balances at 5,488.75) and a tool list of 71 read tools with no create, update or delete.
2. Skill	`/monthly-analysis <client> <YYYY-MM>`: an intake gate, the engine (MoM, TTM, margins, OPEX categories, one-offs, cash runway, gross-margin-drop flags), numbered commentary, a `finance-reviewer` agent, and an Excel sign-off pack with blank sign-off cells. A second client (`xyz`, different layout) runs from a config file with no engine change.
3. Evals	33 golden cases derived by hand from source cells, a commentary grader, a mutation test (19 injected bugs), a 38-fault robustness suite and 186 unit tests.
4. Architecture	A one-page note (`04_architecture/architecture.md`), a workflow page for each of seven workflows, and a traceability matrix from every line of the brief to a command or file.
Extra	A Streamlit review workspace (`05_review_workspace/`) that only displays numbers read from JSON files in `outputs/`, plus approve, reject and lock with an audit log.

Deliverables map

Brief item	Where	PDF in this folder
Working connection	`01_connection/`	01_Connection_Architecture
Skill files	`.claude/skills/`, `.claude/agents/`, `.claude/commands/`, `02_skill/`	02_Skill_and_Analysis_Workflow
Eval cases and grading	`03_evals/`	03_Evals_and_Regression
One-page architecture note	`04_architecture/architecture.md`	04_Architecture_OnePage
Workflow documentation	`04_architecture/workflows/`	05_Workflows_Appendix
Proof of every brief line	`04_architecture/verification_matrix.md`, `03_evals/robustness/`	06_Verification_and_Robustness_Report

Results (from `03_evals/FINAL_RUN.md`, 2026-10-22)

Check	Result
Unit tests	186 run, 186 OK
Golden cases	33/33
Mutation test	19/19 caught

Check	Result
Robustness (fault injection)	38/38 pass, 0 SILENT-WRONG (6 stricter than declared), source workbook hash unchanged (450 s)
Secret scan of tracked files	0 findings, 230 text files (250 tracked paths)
Source workbook SHA-256	unchanged (`75da8d679a89551f…`) before and after every run

A flawed-commentary control is also graded and **fails** as intended (11/12 numbers, 2/3 movements, cash risk missing), which shows the grader can fail.

What is not done

- **Stretch goal: not done.** The workbook was never loaded into the sandbox. `load_workbook_to_sandbox.py` produces an 18-entry dry-run plan only; a live P&L check found no `Workbook Load` accounts, and `--compare` exits 1.
- **Team hosting is a design, not a build.** Per-client OAuth grants, token stores and server processes are labelled *Proposed*. The server has a stdio transport only.
- **Ten clients are untested.** Two clients exist (`abc`, `xyz`). Isolation is an in-process audit hook plus name-based hooks, not an OS or credential boundary.
- **The skill was not re-run live in the final run.** The saved skill output is graded as it stands on disk.
- **Tone and causes are not code-graded.** The reviewer is an LLM and varies between runs; a human decides.
- **QuickBooks data is not an input to the analysis.** The sandbox holds Intuit's sample company; the analysis reads the supplied workbook.
- **The MCP server is an untracked, git-ignored clone**, so the connection check is not reproducible from the repo alone.
- **Only wide-sheet workbooks** (one column per month) are supported without code changes.